

Day 9 Of Communication System Class

• Problem w/ QAM (PSK)
- phase change $\rightarrow$ distortion
- smooth transition भयन

[MSK - Multiple SK
QSK - Gaussian SK] आर्के
पहने
□: slides

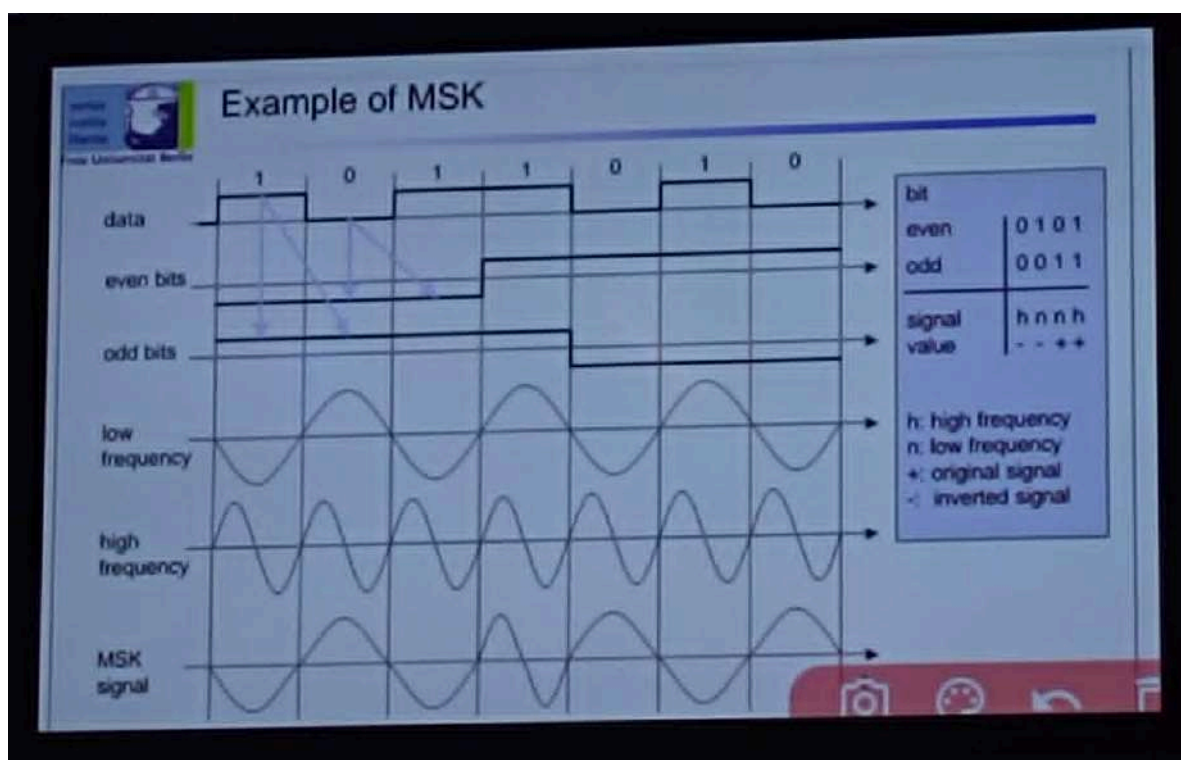
[Most Probable Questions]

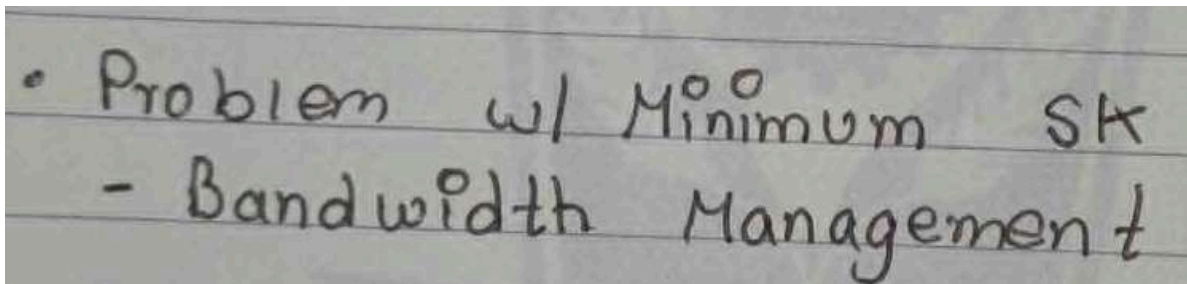
MSK - Example [Exam]

Minimum Shift Keying (MSK)

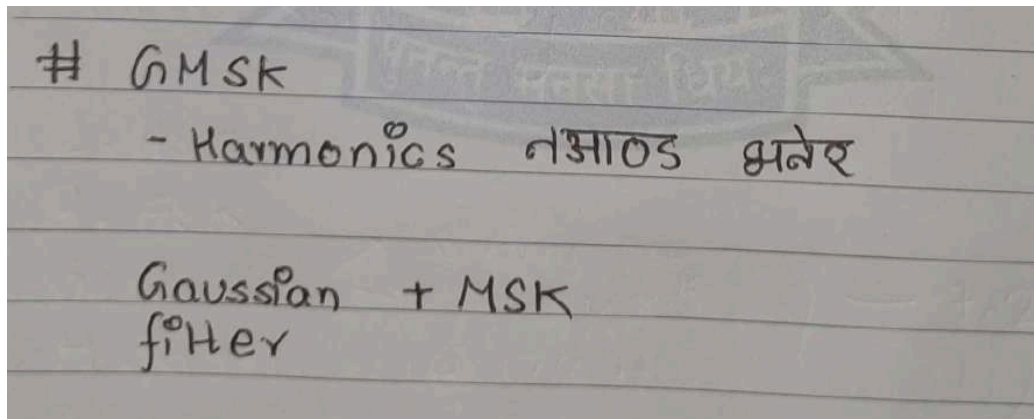
MSK - A special case of binary continuous phase FSK modulation technique in which the change in carrier frequency from symbol 0 to symbol 1 or vice versa is exactly equal to one-half the bit rate of input data signal.

$$v_{\text{MSK}}(t) = \begin{cases} \sqrt{\frac{2E_b}{T_b}} \cos [2\pi f_1 t + \theta(0)] & \text{for binary 1} \\ \sqrt{\frac{2E_b}{T_b}} \cos [2\pi f_2 t + \theta(0)] & \text{for binary 0} \end{cases}$$



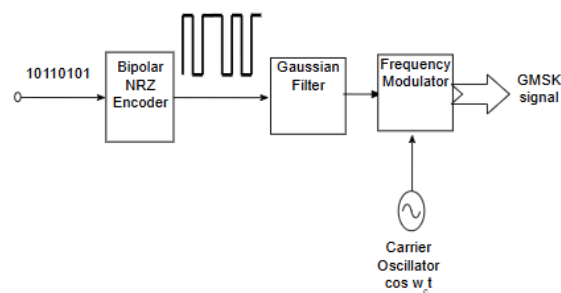


• Problem w/ Minimum SK
- Bandwidth Management

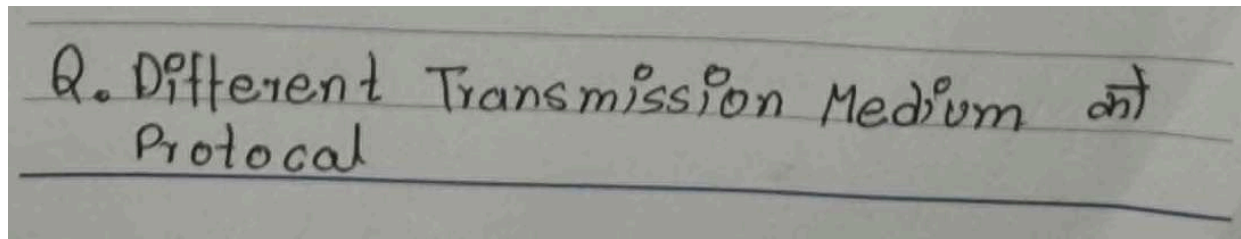


Gaussian Minimum Shift Keying (GMSK)

GMSK - A special case of MSK in which a pre-modulation low-pass Gaussian filter is used as a pulse shaping filter to reduce the bandwidth of the baseband signal before it is applied to MSK modulator.



[Most Probable Questions]

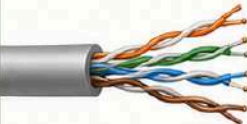
**Q. Write About Different Transmission Mediums With Their Protocols .**

Transmission Medium	Core Type	Primary Protocol(s)	Typical Use Case
Twisted Pair	Wired	Ethernet (802.3), PPPoE	Office LANs, DSL Internet
Coaxial Cable	Wired	DOCSIS	Cable Internet, TV Networks
Fiber Optic	Wired	GPON, SONET/SDH, Fibre Channel	High-speed Internet Backbones, FTTH
Radio Waves	Wireless	Wi-Fi (802.11), Bluetooth, 5G	Mobile networks, Smart home devices
Microwaves	Wireless	DVB-S2, WiMAX	Satellite communication, Long-range towers
Infrared	Wireless	IrDA	Remote controls, short-range peripherals

DIFFERENT TRANSMISSION MEDIUMS • WITH THEIR PROTOCOLS •

A transmission medium is the physical path through which data travels from sender to receiver. Depending on the type of signal and medium used, data transmission can be guided (wired) or unguided (wireless).



TYPE OF MEDIUM	DESCRIPTION & EXAMPLES	COMMON PROTOCOLS USED
GUIDED (WIRED) MEDIA		
1. TWISTED PAIR CABLE 	- Consists of pairs of insulated copper wires twisted together. - Cheap, easy to install and maintain. - Used in telephone lines and LANs. Examples: UTP (Unshielded Twisted Pair), STP (Shielded Twisted Pair)	- Ethernet (IEEE 802.3) – LAN communication - Token Ring (IEEE 802.5) – (legacy) - DSL (Digital Subscriber Line) – Internet over phone lines - HDLCL (High-level Data Link Control) – Reliable data link communication
2. COAXIAL CABLE 	- Consists of a central conductor surrounded by insulation, a metallic shield and outer cover. - Better noise immunity than twisted pair. - Used in cable TV and broadband networks. Examples: TV cables, Cable Internet	- DOCSIS (Data Over Cable Service Interface Specification) – Cable Internet - Ethernet (IEEE 802.3) – Over coax in some legacy networks - ISDN (Integrated Services Digital Network) – Over coaxial in older setups
3. OPTICAL FIBER CABLE 	- Uses light pulses to transmit data through thin glass/plastic fibers. - Very high speed and long distance. - Immune to electromagnetic interference. Examples: Single-mode fiber, Multi-mode fiber	- Ethernet (IEEE 802.3z/802.3ae/802.3bs) – High-speed LAN/WAN - SONET/SDH – Telecom backbone networks - Fibre Channel – Storage Area Networks (SAN) - OSPF, TCP/IP, MPLS – Operate over fiber networks
UNGUIDED (WIRELESS) MEDIA		
4. RADIO WAVES 	- Omnidirectional signals that can penetrate buildings and travel long distances. - Used in broadcast radio, TV, mobile communication. Examples: AM/FM radio, TV broadcast, Mobile networks	- Wi-Fi (IEEE 802.11 a/b/g/n/ac/ax) – WLANs - Bluetooth (IEEE 802.15.1) – Short-range communication - ZigBee (IEEE 802.15.4) – IoT and sensor networks - GSM/3G/4G/5G (Cellular Protocols) – Mobile communication
5. MICROWAVES 	- High-frequency radio waves with directional propagation. - Used for point-to-point communication, satellite links. Examples: Satellite communication, Wireless backhaul	- WiMAX (IEEE 802.16) – Metropolitan area networks - Microwave Communication Links – Point-to-point links - Satellite Communication Protocols – DVB-S/S2, CCSDS
6. INFRARED WAVES 	- Short-range, line-of-sight communication using infrared light. - Cannot penetrate walls. - Used in remotes, IrDA devices. Examples: TV remote, IrDA devices	- IrDA (Infrared Data Association) – Wireless short-range data transfer - HP IrDA, ASK IR Protocols – Device-specific implementations



Key Takeaway: The choice of transmission medium depends on factors like cost, speed, distance, bandwidth, security and environment. Protocols define the rules and formats that ensure effective and reliable communication over these mediums.